

Function Calls

Learning Objectives

- Distinguish between function definition and function invocation
- Pass arguments to functions and capture return values
- Trace the execution flow of a program with multiple function calls

Engage (5-7 minutes)

Think of a function like a recipe in a cookbook. Writing the recipe is defining it, but actually cooking the dish is calling it. A single recipe can be used to cook for different numbers of people by changing the quantities. Today we learn how to invoke functions, pass them different inputs, and use their results in larger programs.

Explore (10-12 minutes)

Consider this scenario: you have a function that adds two numbers. Write code that calls this function with the values 5 and 3, stores the result, then calls it again with 10 and 7. Compare the outputs. Now try calling it without providing any arguments and observe what happens. Discuss the error messages you receive with a partner.

Explain (10-12 minutes)

A function call (or invocation) executes the code inside the function. You call a function by writing its name followed by parentheses, optionally containing arguments. Arguments are the actual values you pass, while parameters are the variable names in the function definition. The return value is what the function sends back, which you can store in a variable for later use.

Elaborate (8-10 minutes)

Build a program with three functions: `add`, `multiply`, and `power`. Call each function multiple times and store results in variables. Then chain the calls—use the return value of `add` as an argument to `multiply`. Trace through your code step by step and write down what value each variable holds at every line. Experiment with passing the result of one function directly as an argument to another without storing it first.

Evaluate (5-8 minutes)

Given a function `def calculate_bmi(weight, height): return weight / (height ** 2)`, write three different function calls with different values and predict the output before running the code. Explain the difference between a function definition and a function call in your own words.